

Obs	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
2	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
3	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
4	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
5	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
6	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
7	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
8	44	1	1	120	263	0	1	173	0	0	2	0	3	1
9	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
10	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
age	303	54.3663366	9.0821010	29.0000000	77.0000000
sex	303	0.6831683	0.4660108	0	1.0000000
cp	303	0.9669967	1.0320525	0	3.0000000
trestbps	303	131.6237624	17.5381428	94.0000000	200.0000000
chol	303	246.2640264	51.8307510	126.0000000	564.0000000
fb	303	0.1485149	0.3561979	0	1.0000000
restecg	303	0.5280528	0.5258596	0	2.0000000
thalach	303	149.6468647	22.9051611	71.0000000	202.0000000
exang	303	0.3267327	0.4697945	0	1.0000000
oldpeak	303	1.0396040	1.1610750	0	6.2000000
slope	303	1.3993399	0.6162261	0	2.0000000
ca	303	0.7293729	1.0226064	0	4.0000000
thal	303	2.3135314	0.6122765	0	3.0000000
target	303	0.5445545	0.4988348	0	1.0000000

Obs	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	0.9506240215	0.6798805249	1.9698642473	0.7626940758	-0.255910365	2.3904835162	-1.004170712	0.0154172814	-0.695480041	1.0855422911	-2.270822075	-0.713248971	-2.145323783	0.9130188171
2	-1.912149695	0.6798805249	1.0009212815	-0.092584625	0.0720802521	-0.416944799	0.8974775738	1.6307737425	-0.695480041	2.1190672376	-2.270822075	-0.713248971	-0.512074772	0.9130188171
3	-1.471722969	-1.465992382	0.0319783157	-0.092584625	-0.815423771	-0.416944799	-1.004170712	0.9758995015	-0.695480041	0.3103985813	0.9747396642	-0.713248971	-0.512074772	0.9130188171
4	0.1798772518	0.6798805249	0.0319783157	-0.662770426	-0.198029668	-0.416944799	0.8974775738	1.2378491979	-0.695480041	-0.206363892	0.9747396642	-0.713248971	-0.512074772	0.9130188171
5	0.2899839332	-1.465992382	-0.93696465	-0.662770426	-0.198029668	-0.416944799	0.8974775738	0.5829749569	1.4331103867	-0.37861805	0.9747396642	-0.713248971	-0.512074772	0.9130188171
6	0.2899839332	0.6798805249	-0.93696465	0.4776011755	-1.046946559	-0.416944799	0.8974775738	-0.071899284	-0.695480041	-0.550872207	-0.648041205	-0.713248971	-2.145323783	0.9130188171
7	0.1798772518	-1.465992382	0.0319783157	0.4776011755	0.9209971433	-0.416944799	-1.004170712	0.1463921296	-0.695480041	0.2242715024	-0.648041205	-0.713248971	-0.512074772	0.9130188171
8	-1.141402925	0.6798805249	0.0319783157	-0.662770426	0.3228966063	-0.416944799	0.8974775738	1.0195577842	-0.695480041	-0.895380523	0.9747396642	-0.713248971	1.1211742386	0.9130188171
9	-0.260549474	0.6798805249	1.0009212815	2.3021957372	-0.911891599	2.3904835162	0.8974775738	0.5393166742	-0.695480041	-0.464745129	0.9747396642	-0.713248971	1.1211742386	0.9130188171
10	0.2899839332	0.6798805249	1.0009212815	1.047786976	-1.509992136	-0.416944799	0.8974775738	1.063216067	-0.695480041	0.482652739	0.9747396642	-0.713248971	-0.512074772	0.9130188171

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=5 Maxiter=1

Initial Seeds		
Cluster	age	chol
1	1.391050747	6.130259885
2	-2.132363057	0.689474355
3	2.161797517	0.438658001
4	0.289983933	-2.320321896
5	0.179877252	3.139757200

Criterion Based on Final Seeds = 0.5873

Cluster Summary						
Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	1	.	0		5	4.5097
2	91	0.6385	1.9684		4	1.4928
3	99	0.5426	1.9239		4	1.3691
4	82	0.5560	2.0192		3	1.3691
5	30	0.6200	1.6911		3	1.5562

Statistics for Variables				
Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
age	1.00000	0.58366	0.663857	1.974926
chol	1.00000	0.58492	0.662402	1.962106
OVER-ALL	1.00000	0.58429	0.663130	1.968502

Pseudo F Statistic = 146.65

Approximate Expected Over-All R-Squared = 0.80543

Cubic Clustering Criterion = -12.369

WARNING: The two values above are invalid for correlated variables.

Cluster Means		
Cluster	age	chol

Cluster Means		
Cluster	age	chol
1	1.391050747	6.130259885
2	-1.154712524	-0.075907757
3	0.849414850	0.233249735
4	0.087226508	-0.904127115
5	0.414771505	1.727468190

Cluster Standard Deviations		
Cluster	age	chol
1	.	.
2	0.5463173455	0.7190060477
3	0.5797939882	0.5026789304
4	0.6166919585	0.4878372922
5	0.6132071513	0.6266267304

